

Interpreting PPA and ALM

$$\min: f(x)$$

$$\text{Prox Point Alg: } x_{t+1} \leftarrow \text{prox}_{\lambda f}(x_t)$$

$$x_{t+1} = (I + \lambda \partial f)^{-1}(x_t)$$

$$\text{Primal: } \min: f(x) + g(Ax)$$

$$\text{Dual: } \max: -f^*(-A^T z) - g^*(z)$$

PPA in Dual is Augmented Lagrangian Method (ALM) in primal.

$$z_{t+1} = z_t + \lambda (A\hat{x} - \hat{y})$$

$$(\hat{x}, \hat{y}) = \arg \min: f(x) + g(y) + z_t^T (Ax - y) + \frac{\lambda}{2} \|Ax - y\|_2^2.$$

Moreau - Yosida Smoothing

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f_{(\lambda)}(x) \triangleq \inf_u \left(f(u) + \frac{1}{2\lambda} \|u - x\|_2^2 \right)$$

$$= \text{infimal convolution of } f \text{ and } \|\cdot\|_2^2 / 2\lambda$$

$$= \inf_{u+v=x} f(u) + \frac{1}{2\lambda} \|v\|_2^2$$

$$\text{Also : } f_{(\lambda)}(x) = f(\text{prox}_{\lambda f}(x)) + \frac{1}{2\lambda} \|\text{prox}_{\lambda f}(x) - x\|_2^2$$

Properties of the Moreau Envelope

Conjugate of Moreau Envelope

Recall: conjugate maps inf. conv to sum

$$\text{i.e. } F(x) = f_1 \square f_2(x)$$

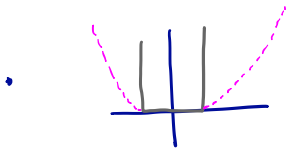
$$\Rightarrow F^*(y) = f_1^*(y) + f_2^*(y)$$

$$\text{Thus: } (f_{(\lambda)})^*(y) = f^*(y) + \frac{\lambda}{2} \|y\|_2^2$$

Properties: • $f_{(\lambda)}(x)$ is convex because it is the infimum over u of a fn $F(x, u) = f(x) + \frac{1}{2\lambda} \|x - u\|_2^2$ jointly convex in x, u (Exercise: show this)

• $f_{(\lambda)}(x)$ is smooth.

Examples



$$f(x) = |x|$$

$$\Rightarrow f_{(\lambda)}(x) = \frac{1}{2\lambda} \cdot \text{dist}(x, \mathcal{X})^2$$

$$f_{(\lambda)}(x) = \inf_w f(w) + \frac{1}{2\lambda} \|x - w\|_2^2$$

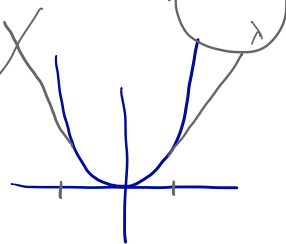
$$= \inf_w \sum (|x_i| + \frac{1}{2\lambda} (x_i - w_i)^2)$$

$$\cdot f(x) = \sum |x_i| = \|x\|_1$$

$$\Rightarrow f_{(\lambda)}(x) = \sum_i h_{\lambda}(x_i)$$

$$\text{where: } h_{\lambda}(z) = \begin{cases} z^2/2\lambda & |z| \leq \lambda \\ |z| - \lambda/2 & |z| \geq \lambda \end{cases}$$

$$\min: |z| + \frac{1}{2\lambda} (z - w)^2$$



$$0 \in \partial |z| + \frac{1}{\lambda} (z - w)$$

Gradient of Moreau Envelope

$$f_{(\lambda)}(x) = \inf_u: f(u) + \frac{1}{2\lambda} \|u - x\|_2^2$$

$f_{(\lambda)}$ is convex and smooth.

If f is closed $\Rightarrow f_{(\lambda)}$ is closed.

$$\Rightarrow f_{(\lambda)}(x) = (f_{(\lambda)})^{**}(x) = \sup_y \left\{ x^T y - (f^*(y) + \frac{\lambda}{2} \|y\|_2^2) \right\}$$

$$x - \lambda y \in \partial f^*(y) \iff y \in \partial f(x - \lambda y)$$

maximizer in dual is unique by strong convexity

$$\hat{y} = \underbrace{\nabla f_{(\lambda)}(\hat{x})}_{\substack{\uparrow \\ \text{check!}}} = \frac{1}{\lambda} (\hat{x} - \text{prox}_{\lambda f}(\hat{x})) = \underset{\substack{\uparrow \\ \text{Moreau decomposition}}}{\text{prox}_{\frac{1}{\lambda} f^*}(\hat{x}/\lambda)}$$

Gradient Descent on Moreau Envelope and PPA

$$f_{(\lambda)}(x) = \inf_u : f(u) + \frac{1}{2\lambda} \|u - x\|_2^2$$

$$\begin{aligned} \text{GD: } x_{t+1} &= x_t - \lambda \nabla f_{(\lambda)}(x_t) \\ &= x_t - \lambda \left(\frac{1}{\lambda} (x_t - \text{prox}_{\lambda f}(x_t)) \right) \\ &= \text{prox}_{\lambda f}(x_t) \end{aligned}$$

- (A) PPA is Gradient Descent on a smoothed version of f
- (B) Minimizer of Moreau Envelope = minimizer of our original function.

Back to Augmented Lagrangian

$$\begin{aligned} (\hat{x}, \hat{y}) &= \underset{x, y}{\operatorname{argmin}} : f(x) + g(y) + \frac{\lambda}{2} \|Ax - y + \frac{z}{\lambda}\|_2^2 \\ z_{t+1} &= z_t + \lambda (A\hat{x} - \hat{y}) \end{aligned}$$

ALM is PPA in dual = GA on a smoothed dual

can eliminate y in the primal: smoothed g

$$\hat{x} = \underset{x}{\operatorname{argmin}} : f(x) + g_{(\frac{1}{\lambda})}(Ax + \frac{z}{\lambda})$$

Ex: $f(x) + \|Ax - b\|_1$

Huber loss

$$\hat{x} = \underset{x}{\operatorname{argmin}} : f(x) + h_{(\frac{1}{\lambda})}(Ax - b + \frac{z}{\lambda})$$